Master's Thesis

Teil 3/4 - R&D Design and Work Plan

RAG ingestion framework: Standardizing the heterogeneity of structured and unstructured enterprise data

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Abstract

The third part outlines the research and development design and work plan for a standardized RAG ingestion architecture that addresses the theoretical findings of the Knowledge Base. The artifact's development is based on the DSR framework by Hevner et al. (2004), and is accomplished in four iterations. The artifact uses large language models to autonomously address terminological heterogeneity across structured and unstructured sources by mapping financial labels to a Unified Document Model. A rigorous validation is ensured by a multi-method evaluation strategy. This comprises a controlled experiment to benchmark engineering effort reduction, a semantic mapping evaluation to assess terminology harmonization precision and recall, and a faithfulness assessment to validate that answers are grounded in source documents. The artifact is considered successfully validated only if all three evaluations are validated simultaneously. A work plan, which is illustrated in a Gantt chart, structures the build and evaluation cycles in Part 4.

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1 Introduction

The Knowledge Base revealed that traditional ingestion models cannot efficiently process and standardize heterogeneous data. Furthermore, a critical gap in existing literature was revealed that current research lacks standardized semantic harmonization frameworks designed specifically for ambiguous enterprise documents, such as annual financial reports. The artifact autonomously ingests heterogenous annual financial reports from Austrian companies provided by Compass Gruppe GmbH (2026), dynamically maps ambiguous terminology to a Unified Document Model, and preserves contextual richness to prevent LLM hallucinations. Part 3 transitions into the build and evaluate cycles of the DSR framework, as defined by Hevner et al. (2004). This part uses a four-iteration development process and rigorous evaluation methods structured across a work plan in a Gantt chart.

2 Artifact Design and Iterations

The development of this artifact follows a rigorous, iterative approach designed to validate the RAG ingestion architecture. It is conceptually divided into four developing iterations that address the functional and non-functional requirements established in the knowledge base. The architecture diagram of the artifact is illustrated in Figure 1.

2.1 Iteration 1: Architectural Setup

The first iteration establishes the artifact's foundational, modular structure to ensure reproducibility and scalability. LangChain serves as the orchestration framework, seamlessly coordinating components and eliminating the need to program LLM interactions from scratch. The artifact preliminarily uses GPT-4o-mini, though the final model will be selected based on operational costs, extraction accuracy, and output quality. The core deliverable of this phase is the Unified Document Model (UDM), which is a universal template that aligns diverse inputs, such as unstructured PDFs and structured JSONs, into a single, standardized format.

2.2 Iteration 2: Data Ingestion Pipelines

The second iteration develops a hybrid data ingestion pipeline to convert heterogeneous financial reports into the standardized UDM. Supported by LangChain's document loaders, the architecture automatically routes structured JSON files through a systematic parsing pipeline to extract all financial details. Simultaneously, unstructured PDFs are processed using optical character recognition (OCR) to reliably extract machine-readable text. This automated routing ensures that both data formats are seamlessly unified into a single UDM.

2.3 Iteration 3: Semantic Harmonization and Chunking

The third iteration resolves the challenge of ambiguous financial terms (e.g., "Umsatzerlöse" and "Revenue"). An LLM semantic mapper simulates human domain knowledge to harmonize these synonymous terms into a single standard. LangChain supports to process specific instructions, ensuring context-aware translations like mapping "Abschreibungen" to either "depreciation" or "amortization". LangChain's Text Splitter prepares the unstructured data via semantic chunking. Dividing text contextually into smaller units, rather than cutting paragraphs randomly, prevents the fragmentation of critical financial details. This iteration delivers a standardized dataset that prepares and structures the data for the retrieval process.

2.4 Iteration 4: Retrieval Process

In the fourth iteration, the RAG pipeline is constructed to retrieve insights from the financial reports. Text chunks from the third iteration are vectorized to enable semantic searches based on conceptual similarity rather than exact keyword matches. LangChain's retriever automates this pipeline, seamlessly coordinating vector searches and the answer generation process. For the evaluation and testing, a minimal user interface is developed that allows the author to submit predefined queries directly to the RAG pipeline and inspect the generated answers alongside their source citations. To ensure reliability and prevent LLM hallucinations, the artifact is strictly restricted to formulate answers using only the ingested documents. Every answer explicitly links to its original source document to ensure complete traceability. This iteration delivers an artifact capable of accurately answering complex financial questions using instantly retrieved, context-aware data.

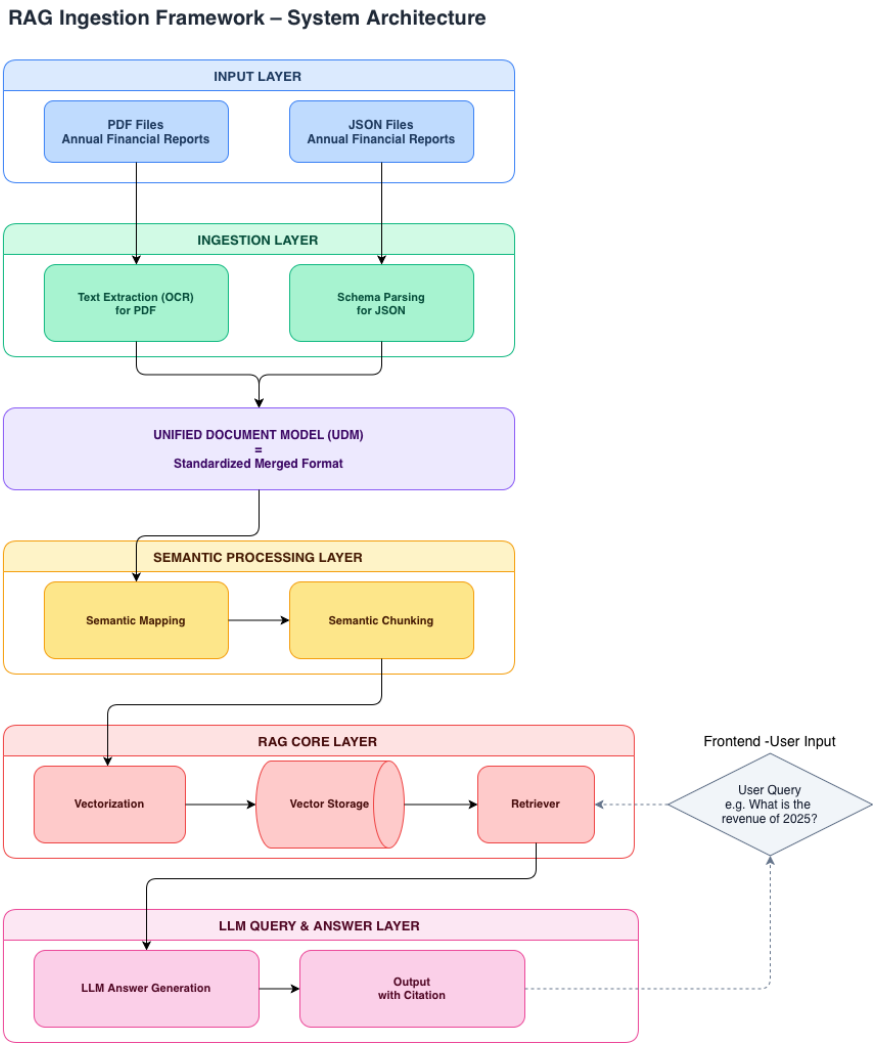

Figure 1: Architectural Diagram of the Artifact

3 Evaluation Design and Work Plan

To rigorously demonstrate the utility and quality of the design artifact, this approach validates the three research properties of reducing manual engineering effort, reliability and contextual richness. The correlation between requirements, components, metrics and threshold is illustrated within a Matrix in Table 1. Within this industry a balance between performance and quality is required to determine the artifact's success. Therefore, the artifact is considered successfully validated only if it reaches the predefined thresholds across all three evaluations. A sample of eight annual financial reports was selected from the Compass Gruppe GmbH (2026) to represent the two data types across four real Austrian companies.

3.1 Controlled Experiment

This method validates the artifact's primary value proposition of reducing manual engineering effort. Following the work of Venable et al. (2016), who emphasize artificial evaluation in controlled settings to assess functional performance before organizational deployment, this experiment compares the artifact to the current manual baseline.

- **Test Setting:** Due to limitations in organizational resources, the manual baseline is measured retrospectively via a structured survey completed by the data integration team lead, while artifact runtime is measured directly by the author.
- **Test Scenario A - Manual Engineering Effort (Baseline):** The data integration team lead estimates the required time to parse and ingest heterogenous annual reports from four companies into the organizational database, covering four tasks: document analysis, script development, debugging and validation, and data upload. The preliminary manual engineering workflow, which will be validated by the team lead, is defined in the Appendix A, Figure A1. The baseline is estimated between 6 and 8 hours.
- **Test Scenario B - Artifact Performance:** The artifact processes the same documents, by uploading the documents, configuring settings, automated processing, reviewing flagged mappings with low confidence, validating outputs with a human reviewer. The architecture of the artifact is illustrated in Figure 1. The time is estimated between 70 and 90 minutes.
- **Success Criterion:** The artifact only passes if the criterion is successful
    - **Time Reduction =** (Manual Time – Artifact Time) / Manual x 100
      Threshold: >75% must be saved by the artifact. The threshold was defined to guarantee a substantial productivity enhancement over the manual baseline. A lower threshold would not justify the adoption costs and architectural complexity of the framework.

3.2 Static Benchmarking Analysis

This method evaluates the artifact's quality, reliability and semantic precision. According to Yu et al. (2025), the standard methodology for objectively measuring component-level performance in RAG systems is static benchmarking against a ground truth dataset.

3.2.1 Semantic Mapping Evaluation

This methodology employs the static benchmarking analysis to validate semantic mapping. It refers to the "Valentine" framework by Koutras et al. (2021), a schema matching benchmark that requires the use of an expert ground truth to objectively measure automated mapping performance. As each corporate financial report has different terminology, there is no universal database to standardize these terms, and a ground truth is established by experts.

- **Test Scenario:** A standard dataset of 50 financial terms from the sample reports is defined by the author. Each source term is paired with its correct target term (e.g., "Umsatzerlöse" with "revenue") to build the ground truth. The sample size of 50 terms was selected to provide a sufficiently representative baseline for assessing the system's accuracy. The LLM semantic mapper then processes the same 50 terms, and its predictions are compared to the standard dataset.
- **Success Criteria:** The artifact only passes if both criteria are successful
    - **Precision:** True Positives / (True Positives + False Positives)
      Threshold: >85% Precision was defined, because a 15% incorrect mapping rate would have a significant impact on the retrieval process, as a term mapped to the wrong standardized term would cause incorrect financial figures within the generated answers, which is unacceptable in financial reporting.
    - **Recall:** True Positives / (True Positives + False Negatives)
      Threshold: >80% Recall is acceptable because any undetected terms are flagged for the manual human review. This ensures that final output quality is never compromised, even if the system misses up to 20% of the mappings.

3.2.2 Faithfulness Score Assessment

This method measures the reliability of the artifact by measuring faithfulness, defined as the extent to which generated responses are grounded in source documents to mitigate LLM hallucinations. According to Malin et al. (2025), faithfulness is the definitive metric for assessing generative LLM reliability in retrieval-augmented generation architectures, as it quantifies the alignment between model outputs and verifiable source material.

- **Test Scenario:** A dataset of 30 predefined questions (e.g. What was the revenue of 2025?) is created by the author, and each generated answer is manually validated against the source documents. An answer is marked as "faithful" only if all facts are explicitly present in the ingested source documents.
- **Success Criterion:** The artifact only passes if the criterion is successful
    - **Faithfulness Score =** (Number of Faithful Answers / Total Answers) x 100 **Threshold: >85%** Unfaithful answers referencing an incorrect figure could lead to a misleading business decision.

Requirements and Evaluation Matrix							
Nr.	**Requirements**	**Iteration**	**Component**	**Evaluation Method**	**Metric**	**Threshold**	**Reasoning**
FR1	Multi-Source Ingestion	2	Hybrid Ingestion Pipeline	Controlled Experiment	Time Reduction	>75%	A reduction of at least 75% ensures the framework delivers an unambiguous productivity gain over manual ingestion. A lower threshold would not sufficiently justify the architectural complexity of a hybrid ingestion pipeline
FR2	Semantic Flattening	3	LLM Flattening Module	Semantic Mapping Evaluation	Precision Recall	>85% >80%	This Precision of ensures that most flattened terms are correctly standardized. This Recall is acceptable because the remaining 20% of unrecognized terms are routed to a human review.
FR3	Semantic Chunking	3	Semantic Chunker	Faithfulness Assessment	Faithfulness Score	>85%	A threshold of >85% ensures chunks preserve sufficient semantic boundaries for reliable answer generation.
FR4	Schema Mapping	3	LLM Semantic Mapper	Semantic Mapping Evaluation	Precision Recall	>85% >80%	This threshold ensures that incorrect schema mappings do not harm the Unified Document Model. An artifact that maps 15% of terms to the wrong standard term would return incorrect financial figures in generated answers.
FR5	Vectorization	4	Vector Store	Faithfulness Assessment	Faithfulness Score	>85%	The threshold confirms that vectorized chunks retain sufficient context to support grounded answer generation.
FR6	RAG Outcome	4	RAG + LLM	Faithfulness Assessment	Faithfulness Score	>85%	The threshold ensures that speed is only validated alongside verified output reliability.
FR7	Orchestration Integration	all	LangChain	All methods	All metrics	-	An own threshold is not applicable as this is a supporting infrastructure component.
FR8	Metadata Enrichment	3	Metadata Enricher	Faithfulness Assessment	Faithfulness Score	>85%	Metadata tags such as company name and document type are required for faithful sources in generated answers.
NFR1	Contextual Richness	4	RAG Pipeline	Faithfulness Assessment	Faithfulness Score	>85%	The threshold ensures the pipeline limits answer generation strictly to retrieved content without hallucination.
NFR2	Interpretability & Traceability	4	Answer Synthesizer	Faithfulness Assessment	Faithfulness Score	>85%	The threshold ensures every cited source supports the generated answer.
NFR3	Standardization	1	Architectural Setup	Controlled Experiment	Time Reduction	>75%	The time reduction threshold validates standardization by demonstrating that the artifact processes all documents within the hybrid ingestion pipeline without requiring format-specific manual engineering intervention.

Table 1: Evaluation and Validation Matrix

3.3 Work Plan

According to Chapter 2 and 3 the development of the artifact follows a rigorous, iterative approach designed to validate the framework in distinct iterations. This ensures that the core complexities of data heterogeneity are addressed systematically.

3.3.1 Gantt Chart

As illustrated in Table 2, a Gantt chart visualizes the time-based dependencies between the four development iterations and the evaluation phase across sixteen weeks.

Each iteration concludes with a dedicated testing cycle to validate the developed components before the next iteration begins. Iterations intentionally overlap to ensure a continuous development flow, as later iterations begin while earlier ones are completing their testing cycles. The total estimated development effort amounts to 465 hours, covering all four iterations, evaluation, and continuous documentation. A two-week buffer in weeks 15 and 16 is allocated to address unexpected technical challenges before the final prototype is released.

Gantt Chart	Time effort	March				April				May				June			
		Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13	Week 14	Week 15	Week 16
Iteration 1: Architectural Setup	**70 h**																
Project Setup	10 h																
Orchestration Setup	20 h																
UDM Schema Definition	25 h																
Testing Cycle	15 h																
Iteration 2: Data Ingestion Pipelines	**95 h**																
PDF + JSON Loader	15 h																
OCR Extraction for PDF	25 h																
Schema Parsing for JSON	25 h																
Metadata enrichment	15 h																
Testing Cycle	15 h																
Iteration 3: Semantic Harmonization	**95 h**																
Semantic Mapping	40 h																
Semantic Chunking	35 h																
Testing Cycle	20 h																
Iteration 4: Retrieval Process	**125 h**																
Vectorization	20 h																
Vector Storage	15 h																
Retriever & Generation	40 h																
Frontend development	30 h																
Testing Cycle	20 h																
Evaluation	**80 h**																
Controlled Experiment	35 h																
Semantic Mapping Evaluation	25 h																
Faithfullness Assessment	20 h																
Documentation	**80 h**																
Final Prototype	**465 h**																**Release**

Table 2: Gantt Chart - Work Plan

Conclusion

In the third part, the theoretical findings from the knowledge base were operationalized into a concrete research and development design and work plan. The artifact is divided into four development iterations, progressing from the foundational Unified Document Model through data ingestion, semantic harmonization, and the retrieval output. To validate the artifact's utility against the defined functional and non-functional requirements, three evaluation methods were established. A controlled experiment measures the reduction in manual engineering effort, while the semantic mapping evaluation validates terminology harmonization quality through precision and recall. The faithfulness assessment ensures a reliable source-grounded answer generation. The artifact is considered successfully validated only if all three evaluation thresholds are being met simultaneously. The fourth part will document the development process, execute the evaluation, and present the results in relation to the research question.

Bibliography

Compass-Gruppe GmbH. (2026). *Wirtschafts-Compass* [Company Database]. https://www.compass.at

Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design Science in Information Systems Research. *MIS Quarterly*, *28*(1), 75–106. https://doi.org/10.2307/25148625

Koutras, C., Siachamis, G., Ionescu, A., Psarakis, K., Brons, J., Fragkoulis, M., Lofi, C., Bonifati, A., & Katsifodimos, A. (2021). Valentine: Evaluating Matching Techniques for Dataset Discovery. *2021 IEEE 37th International Conference on Data Engineering (ICDE)*, 468–479. https://doi.org/10.1109/ICDE51399.2021.00047

Malin, B., Kalganova, T., & Boulgouris, N. (2025). A Review of Faithfulness Metrics for Hallucination Assessment in Large Language Models. *IEEE Journal of Selected Topics in Signal Processing*, *19*(7), 1362–1375. https://doi.org/10.1109/JSTSP.2025.3579203

Venable, J., Pries-Heje, J., & Baskerville, R. (2016). FEDS: A Framework for Evaluation in Design Science Research. *European Journal of Information Systems*, *25*(1), 77–89. https://doi.org/10.1057/ejis.2014.36

Yu, H., Gan, A., Zhang, K., Tong, S., Liu, Q., & Liu, Z. (2025). Evaluation of Retrieval-Augmented Generation: A Survey. In W. Zhu, H. Xiong, X. Cheng, L. Cui, Z. Dou, J. Dong, S. Pang, L. Wang, L. Kong, & Z. Chen (Eds.), *Big Data* (Vol. 2301, pp. 102–120). Springer Nature Singapore. https://doi.org/10.1007/978-981-96-1024-2_8

List of Tools Used

AI tool	Purpose	Concerned parts of the work	Notes
Deepl Translator https://www.deepl.com/translator	Translation of text passages	Whole document	
Deepl Write https://www.deepl.com/write	Rephrase and grammar check	Whole document	
Claude https://claude.ai/	Architectural Plan	Artifact Design and Iteration	Technical brainstorming on the architecture of the artifact and the iterations
Gemini http://gemini.google.com/	Literature research	Foundational Theories, Systematic Literature Review	Researching literature accessible via the used databases; Early stage SLR explanation and brainstorming phase

Glossary

Term	Abbreviation	Definition
Artificial Intelligence	AI	Technology that enables computers and machines to simulate human learning (Source: IBM: ”What is AI?)
Application Programming Interface	API	Set of rules and protocols that enables software applications to communicate (Source: IBM: “What is an API”)
Extract Transform Load	ETL	A data integration process that combines, cleans and organizes data from multiple sources into a single, consistent data set for storage in a data warehouse (Source: IBM: What is ETL?)
LangChain	LangChain	LangChain is an open-source orchestration framework for application development using large language models (LLMs). (Source: IBM: What is LangChain)
Large Language Model	LLM	Machine Learning Models that comprehend and generate human language (Source: Cloudfare: “Was ist ein großes Sprachmodel?”)
Natural Language Processing	NLP	A subfield of computer science and artificial intelligence (AI) that uses machine learning to enable computers to understand and communicate with human language (Source: IBM: What is NLP?)
Optical Character Recognition	OCR	Optical character recognition (OCR) is a technology that uses automated data extraction to quickly convert images of text into a machine-readable format. (Source: IBM: What is OCR?)
Retrieval Augmented Generation	RAG	Architecture for optimizing a model by connecting it with external knowledge bases to deliver a higher quality (Source: IBM: “What is RAG?”)

Appendix A: Manual Engineering Workflow

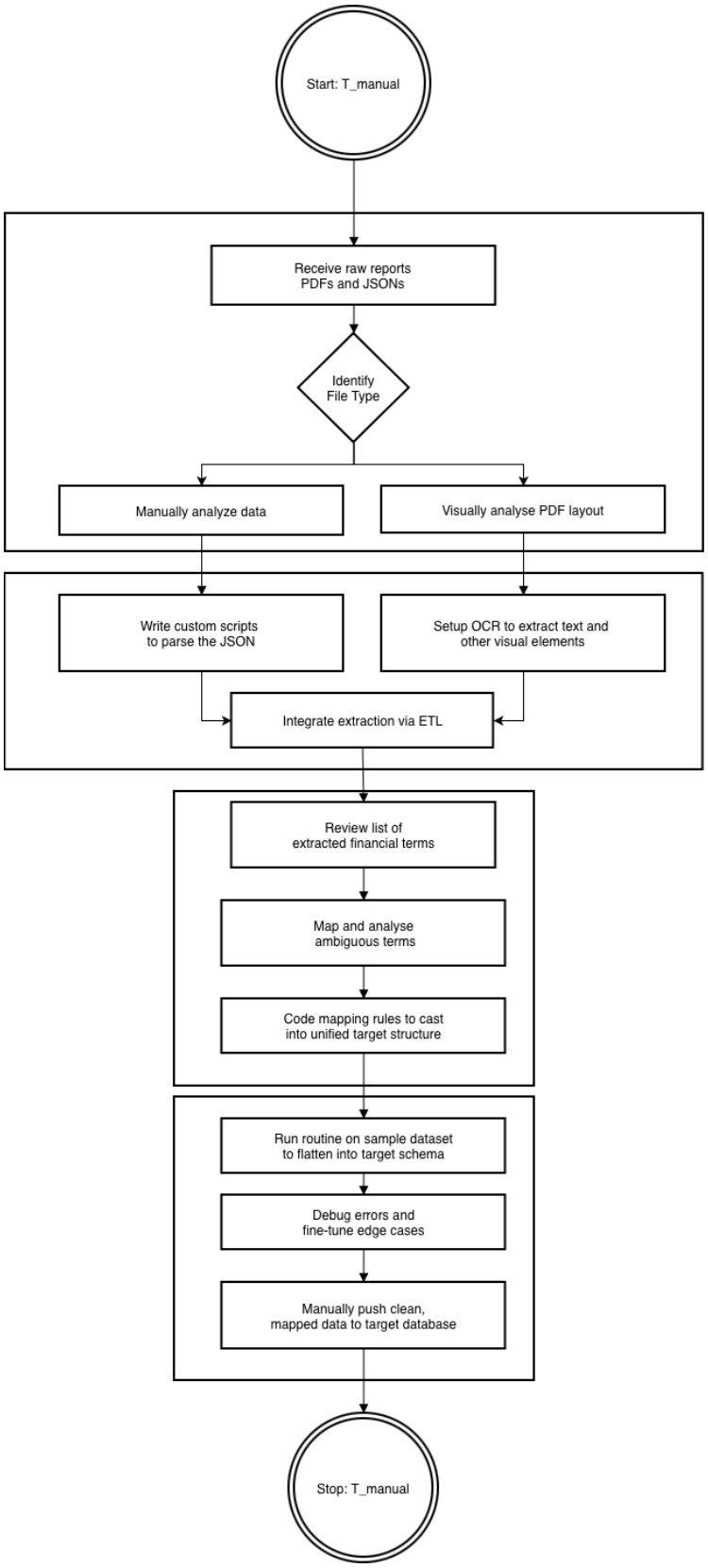

Figure A 1: Manual Engineering Workflow